

CustomerIQ

ML + AI Model Structure

A practical architecture for moving from customer transactions to prediction, risk scoring, business intelligence, and evidence-grounded AI recommendations.

1. Current Foundation

Layer	Current state	Purpose
Raw data	UCI Online Retail II	Original transaction history
Customer dataset	3,705 eligible customers	Model-ready customer representation
Predictors	6 behavioral features	Frequency, Monetary, Recency, Customer Lifespan, Average Purchase Interval, Median Purchase Interval
Target	observed_churn	1 = no observed return; 0 = returned
ML	Days 5–7	Build, compare and validate models
Risk scoring	Day 8	Convert churn probability into actionable tiers
SQL / BI	Day 9	Prepare structured business data for Power BI

2. ML Pipeline

Customer transactions → cleaning → customer-level features → leakage controls → stratified train/test split → model training → validation → churn probability → risk score.

Model	Role	Observed result
Logistic Regression	Interpretable baseline	ROC-AUC 0.7731
Random Forest	Non-linear ensemble	Strong training performance but substantial overfitting concern
Gradient Boosting	Strongest candidate	ROC-AUC 0.7769; F1 0.6990

3. Validation

Day 7 used 3-fold stratified cross-validation plus leakage checks, confusion matrices, ROC and Precision-Recall curves, probability distributions, threshold analysis, and feature-importance analysis.

Leakage controls exclude Customer ID, returned_in_validation, and observed_churn from the predictor matrix.

4. Risk Scoring

Tier	Probability rule	Customers	Share
HIGH	$P(\text{churn}) \geq 0.70$	690	18.62%
MEDIUM	$0.40 \leq P(\text{churn}) < 0.70$	1,511	40.78%
LOW	$P(\text{churn}) < 0.40$	1,504	40.59%

5. Proposed AI Recommendation Architecture

The AI layer should sit on top of the validated ML output. ML answers 'who is at risk?'; AI helps answer 'what could we do next, and why?'.

Stage	Input	Job	Output
1. Customer context	Risk + behavior	Summarize current customer state	Customer profile
2. Pattern detection	Behavior + segment	Identify relevant patterns	Likely drivers
3. Evidence retrieval	Approved historical cases / documents	Retrieve comparable evidence	Evidence set
4. AI reasoning	Prediction + context + evidence	Generate possible actions	Recommendation
5. Guardrails	Recommendation + evidence	Check grounding and uncertainty	Approved suggestion
6. Action	Approved suggestion	Translate into business action	Campaign / retention action
7. Feedback	Action + outcome	Measure result	Learning evidence

6. Evidence-Grounded AI

Your proposed idea fits this architecture: historical business cases can be used as evidence for current recommendations when the current pattern is comparable. The system should never claim that a historical case proves the cause of a current customer's behavior.

Recommended reasoning chain:

Historical pattern → observed problem → intervention → measured outcome → comparable current pattern → ML risk signal → AI recommendation → business action → measured result.

The AI output should clearly distinguish four things: observed data, ML prediction, retrieved evidence, and AI-generated recommendation.

7. Full CustomerIQ Architecture

Data → Features → ML → Churn Probability → Risk Segment → SQL → Power BI → Evidence Retrieval → AI Recommendation → Business Action → Outcome Measurement → Feedback Loop

8. ML vs AI

Component	ML	AI recommendation layer
Main question	Who is likely to churn?	What could the business do next?
Input	Six behavioral predictors	ML output + customer context + evidence
Output	Probability / risk tier	Explanation + recommendation
Evaluation	ROC-AUC, F1, recall, validation stability	Evidence quality, usefulness, factual grounding, business outcomes
Main risk	Leakage / overfitting / weak generalization	Hallucination / unsupported recommendations / weak retrieval

9. Next Build Sequence

1. Power BI: build executive and retention dashboards from the Day 9 dataset.
2. Evidence layer: organize approved historical cases, campaign records and business documents.
3. Retrieval: identify relevant historical cases using structured filters and/or semantic search.
4. Recommendation engine: combine risk, behavior, evidence and business rules.
5. Guardrails: require evidence, label uncertainty, and avoid unsupported causal claims.
6. Web app: show customer profile, risk, evidence, recommendation and action tracking.
7. Feedback loop: record interventions and outcomes for future evaluation.

10. Product Principle

CustomerIQ evolves from a churn prediction project into a decision-support system: predict → explain → recommend → act → measure.

Important status note: the ML, risk-scoring and SQL layers described above are completed project work. The AI recommendation and evidence-retrieval layers in this document are the proposed next-stage architecture.